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1. State the differences between procedural programming and OOP.

Data in procedural programming is global and public while OOP has encapsulation.

2. What is an Object and what is a Class? What is the difference between them?

Class is the template of creating an object while object is an instance created from class.

3. What is an Attribute?

Data.

4. What is a Method?

Function.

5. What is encapsulation? How does it relates to information hiding?

Encapsulation is combining data and behavior in one package like a capsule, only exposing the data based on requests. Data hiding is an approach used in encapsulation to make certain parts of the data private.

6. What is inheritance? How does it relates to polymorphism?

Inheritance is creating a child class where structure of the attribute and methods are inherited from the parents. Polymorphism could be used in Inheritance that sets a reference type to call methods for different objects of the same type at once.

1. What are the 7 basic principles of object orientation? Provide a brief description for each principle.

- Divide-and-Conquer: divide means to divide a program into multiple objects by similar nature for easier management, while conquer means to solve the problem or creating project for each object.
- Encapsulation and Modularity: it means that each object should be a self-contained module with a clear responsibility, and attributes and action necessary to maintain its role.

- **Public Interface:** It means objects should have a clear public interface for it to interact with each other.
- **Information Hiding:** It means the unnecessary details of the program that has nothing to do with the functionality intended for user, should be hidden from another object.
- **Generality:** It means classes in OOP should be designed as general as possible to encourage reuse as it can not only be used in one particular task but that particular kind of task.
- **Extensibility:** It means an object should be designed in a way that the object's behavior can be easily extended to carry out more specialized task.
- **Abstraction:** It means that in OOP, we need to focus more on the core features while the other can be abstract at first.

2. State the benefits of OOP.

OOP allows developers to reuse codes by inheritance, polymorphism practice and class created can even be used in other applications. It is also easier to perform debugging as each classes can be tested independently without the worry of destroying other classes. Moreover, the reused object have already been tested, so it is time saving.